

NELLIS AFB, NV, USA

WMO: 746140

Lat: 36.250N

Lon: 115.033W

Elev: 570

StdP: 94.66

Time Zone: -8.00 (NAP)

Period: 96-19

WBAN: 23112

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-1.1	0.3	-18.8	0.8	10.2	-16.4	1.0	11.1	12.8	10.1	11.7	10.9	1.4	40	0.506

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.6	42.9	19.3	41.7	19.0	40.5	18.6	22.5	34.5	21.9	34.4	21.4	34.5	4.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.2	14.9	27.2	17.7	13.6	28.8	16.4	12.5	30.1	68.8	33.7	66.6	34.2	64.5	34.5	24.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.3	10.9	9.2	DB	-3.6	44.9	1.4	1.2	-4.7	45.8	-5.5	46.5	-6.3	47.2	-7.3	48.1	(4)
(5)			WB	-6.5	23.5	1.2	0.6	-7.4	23.9	-8.1	24.2	-8.7	24.5	-9.6	25.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	20.7	9.3	11.2	15.9	19.5	24.6	30.8	33.8	32.4	28.1	20.4	13.6	8.1		
	DBStd	9.45	2.99	3.58	3.42	3.76	4.07	3.24	2.37	2.42	3.46	3.84	3.74	3.01		
	HDD10.0	159	48	26	1	0	0	0	0	0	0	1	9	74		
	HDD18.3	1090	281	199	91	31	4	0	0	0	0	22	145	317		
	CDD10.0	4061	26	61	183	286	452	625	737	694	542	322	119	15		
	CDD18.3	1949	0	1	15	66	197	375	479	436	292	84	4	0		
	CDH23.3	30162	0	13	210	810	2600	6091	8201	7148	4077	931	81	0		
(13)	CDH26.7	19103	0	0	38	286	1343	4041	5793	4842	2417	337	7	0		
(14)	Wind	WSAvg	3.4	2.6	3.3	3.8	4.5	4.3	3.6	3.4	3.2	3.1	2.6	2.7		
(15) Precipitation	PrecAvg	174	25	35	17	9	3	3	14	12	11	15	9	21		
	PrecMax	334	102	96	119	44	11	20	56	48	60	48	56	95		
	PrecMin	57	1	1	0	0	0	0	0	2	0	0	0	1		
	PrecStd	66	27	28	24	10	3	4	11	11	14	15	12	21		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.0	25.2	30.0	35.0	38.9	44.2	45.1	43.0	40.6	35.2	27.6	19.9		
		MCWB	9.8	11.2	13.1	14.8	16.2	18.4	19.7	19.4	19.2	15.6	12.9	10.5		
	2%	DB	19.0	23.0	27.5	32.2	36.7	42.3	43.6	41.9	38.9	32.2	25.6	17.7		
		MCWB	8.7	10.4	12.4	13.9	15.4	18.2	19.7	19.2	18.5	15.3	12.0	9.1		
	5%	DB	17.5	20.8	25.7	29.9	35.1	40.8	42.2	40.8	37.6	30.2	23.9	16.0		
		MCWB	8.2	9.7	11.4	12.8	14.8	17.3	19.5	19.0	18.0	14.4	11.4	8.3		
	10%	DB	15.8	18.6	23.8	27.9	33.2	39.3	40.9	39.7	36.2	28.3	21.7	14.3		
		MCWB	7.4	9.1	10.6	12.1	14.3	16.8	19.3	19.0	17.5	13.6	10.5	7.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.0	13.0	14.0	15.4	16.9	20.8	23.3	23.2	21.9	19.0	14.3	13.0		
		MCDB	15.3	19.3	25.3	31.6	34.8	39.4	35.0	34.2	33.0	26.8	23.9	16.5		
	2%	WB	11.0	11.8	13.0	14.5	16.1	19.6	22.6	22.5	20.8	17.3	13.0	11.5		
		MCDB	14.9	18.8	24.8	29.5	33.9	38.7	35.0	34.1	33.8	26.7	23.6	14.9		
	5%	WB	10.0	10.9	12.1	13.5	15.5	18.5	22.1	21.9	20.1	16.1	11.9	10.2		
		MCDB	14.8	17.4	23.9	27.5	32.8	37.2	34.7	34.0	32.9	26.2	22.2	14.2		
	10%	WB	8.8	10.0	11.1	12.7	14.9	17.6	21.6	21.3	19.3	15.1	10.9	8.8		
		MCDB	14.1	17.3	22.5	26.2	31.6	37.3	34.9	33.8	32.3	25.6	20.9	13.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.9	11.5	12.7	13.0	12.8	14.0	12.6	13.1	13.5	13.1	13.5	11.0		
		MCDBR	14.8	15.5	15.3	16.6	15.9	16.0	14.8	14.8	14.8	15.3	15.9	13.4		
	5% WB	MCWBR	7.8	7.6	6.6	7.2	6.2	6.0	5.2	5.5	5.9	6.8	8.1	7.5		
		MCDWR	10.7	11.9	14.1	14.2	14.1	14.0	11.1	11.3	11.9	12.1	14.1	10.1		
(40) Clear-Sky Solar Irradiance	taub	taub	0.267	0.272	0.286	0.309	0.327	0.323	0.365	0.355	0.319	0.293	0.270	0.266		
		taud	2.613	2.615	2.551	2.464	2.438	2.458	2.431	2.450	2.530	2.587	2.620	2.622		
	Ebn at Noon	Ebn at Noon	952	983	990	974	954	952	910	916	939	942	939	930		
		Edh at Noon	72	81	95	109	114	112	114	110	96	82	71	67		
(44) All-Sky Solar Radiation	RadAvg		3.05	4.01	5.52	6.92	7.86	8.53	7.54	7.00	6.17	4.62	3.44	2.70		
	RadStd		0.24	0.35	0.26	0.30	0.38	0.34	0.38	0.34	0.25	0.27	0.14	0.20		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (14 neighbors)		+0.49	N/A	-1.57	N/A	N/A	N/A	N/A	-117	+189	+81

Nomenclature: See separate page